

Bayesian Network for Predicting Goal Scoring Probability in Football (xG)

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Abstract

This project studies football goal-probability prediction using Bayesian Networks. The task is related to expected goals (xG), where each shot is assigned a probability of becoming a goal. StatsBomb Open Data was used to extract shot-level information such as location, angle, body part, shot type, pressure, goalkeeper context, and nearby players. Two Bayesian Network models were developed: an Expert Bayesian Network based on football knowledge and a Hill-Climbing Bayesian Network learned from data. Logistic Regression, Random Forest feature analysis, and the StatsBomb xG were used for comparison. The results show that Bayesian Networks can provide interpretable probability estimates and useful scenario-based reasoning.

Introduction

Domain

Expected goals (xG) estimates the probability that a shot will become a goal. A close central shot normally has a higher probability than a long-distance or narrow-angle shot. xG is useful because it evaluates chance quality rather than only counting goals or total shots [2]. Bayesian Networks are suitable for this domain because scoring probability depends on interacting factors such as shot geometry, body part, shot type, defensive pressure, and goalkeeper context.

Bayesian Networks have previously been applied to football at the match-outcome level, such as in *pi-football* [1]. This project instead models goal probability at the shot level:

$$P(\text{Goal} = \text{Yes} \mid \text{shot evidence}).$$

The dataset used was StatsBomb Open Data.

Aim

The aim was to test whether Bayesian Networks can be used as interpretable xG models. The project investigated whether an Expert BN can represent football logic, whether a Hill-Climbing BN can learn useful dependencies from data, and how both models compare with Logistic Regression and the StatsBomb xG. The task is strongly data-dependent because the probabilities are estimated from observed shots. With more shots, seasons, leagues, and richer tracking information, the probability estimates should become more stable, especially for rare situations such as open goals, penalties, and one-on-one shots.

Data and preprocessing

After preprocessing, 108,281 shots were used, with a goal rate of 11.16%. This means that only about one in nine shots became a goal, which makes probability modeling more appropriate than simple goal/non-goal classification.

The raw events included shot location, outcome, play pattern, player position, body part, technique, shot type, and freeze-frame information. The freeze-frame data was used to engineer goalkeeper location, teammate and opponent locations, nearby-player features, goalkeeper-context features, and shot-triangle variables. Shot distance was computed from the shot coordinates to the goal, while shot angle was computed from the geometric opening between the shot location and the goalposts.

The dataset also contained rich categorical context, including 258

teams, 9 play patterns, 25 shooter positions, multiple body parts, several techniques such as volley or header, and 5 shot types including open play and penalty. Time-interval analysis was also performed, but no meaningful association was found between game time and goal rate.

Exploratory results

Exploratory analysis showed that shot location is one of the strongest drivers of goal probability. Goals were concentrated in central and close-range pitch areas, as shown in Figure 1. This supports the importance of spatial variables such as Distance, Angle, and goalkeeper-related context.

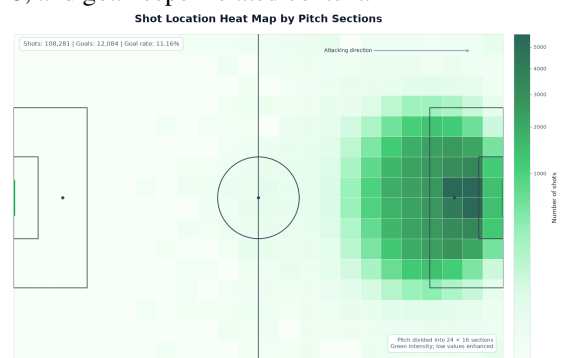


Figure 1: Goal-location heat map by pitch section.

Random Forest feature importance confirmed the same result. The most important variables were related to shot angle, distance to goal, and distance to the goalkeeper. This indicates that goal probability is mainly driven by shot geometry and goalkeeper positioning, while body part, shot type, technique, and pressure provide additional context.

Model

The Expert BN was manually designed from football reasoning. As shown in Figure 2, shot geometry influences GeometryQuality; body part, technique, and first-time shot influence ShotExecution; pressure and nearby opponents influence DefensivePressure; and goalkeeper variables influence GoalkeeperContext. These intermediate concepts, together with contextual variables such as ShotType, PlayPattern, OneOnOne, and OpenGoal, influence Goal. The conditional probability distributions were learned from the data using Bayesian estimation.

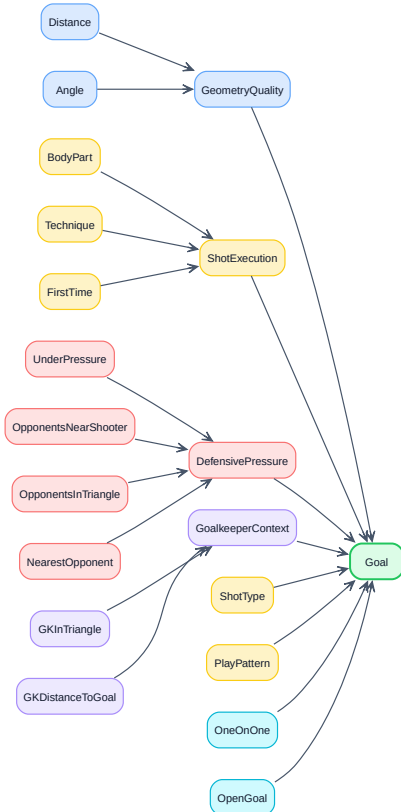


Figure 2: Expert Bayesian Network structure.

The Hill-Climbing BN was learned from data using BIC score. Since the full learned graph contained many dependencies, the Markov blanket of `Goal` was used to summarize the local predictive structure. Figure 3 shows that `Distance`, `BodyPart`, `ShotType`, `OpenGoal`, and `FirstTime` formed the local neighborhood of `Goal`. Once these variables are known, `Goal` is conditionally independent from the rest of the learned network.

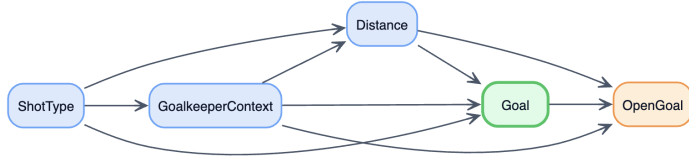


Figure 3: Markov blanket of `Goal` in the Hill-Climbing BN.

Analysis

Experimental setup

The analysis was designed to study both numerical performance and interpretability. Logistic Regression was trained as a statistical xG baseline. The Expert BN was trained using the manually designed structure, while the Hill-Climbing BN was learned from associations in the discretized variables.

The evaluation focused mainly on probability behavior. Brier Score, Log Loss, and MAE measured probability error; ROC AUC and Average Precision measured ranking ability; and total xG measured aggregate calibration. Total xG is the sum of predicted probabilities over all test shots, so it can be a decimal value. Threshold-based metrics such as F1 were also computed, but a fixed 0.5 threshold is naive for football shots because the average goal rate is only around 11%, and most realistic shot probabilities are below 0.5.

The model comparison shows that StatsBomb had the best ranking and probability-error metrics, but it underestimated the total number of goals on this test subset. The learned models were closer in total probability mass to the observed number of goals.

Model	Brier	MAE	AUC	AP	Total xG
StatsBomb	0.0766	0.1494	0.8173	0.4497	2803.37
Log. Reg.	0.0774	0.1537	0.8124	0.4422	2930.18
Expert BN	0.0817	0.1601	0.7720	0.3833	2944.57
Hill BN	0.0822	0.1639	0.7630	0.3397	2949.05

Table 1: Model comparison on 26,960 test shots with 2,944 actual goals.

The Expert BN produced a total xG of 2944.57, almost identical to the 2944 observed goals. The Hill-Climbing BN also produced a close total xG of 2949.05, and Logistic Regression produced 2930.18. This shows that these learned models were well calibrated in aggregate on this dataset. In contrast, StatsBomb xG produced 2803.37 total xG. This does not mean that StatsBomb xG is not probabilistic; rather, it shows that different xG providers and models can produce different probability estimates because they use different data distributions, features, and assumptions. Since shot outcome is uncertain by nature, the goal is not to find a deterministic answer, but to estimate probabilities that match real outcomes as closely as possible over many shots.

Query checks

The most interpretable analysis came from Bayesian Network queries. These queries tested whether the models assigned reasonable probabilities to recognizable football situations before comparing their numerical outputs.

Scenario	Expert BN	Hill BN
Open goal chance	0.8157	0.7372
High-quality close shot	0.4338	0.5070
Header from corner under pressure	0.0852	0.1199
Difficult long-range shot	0.0536	0.0301

Table 2: Goal-probability queries for selected football scenarios.

Both Bayesian Networks followed football intuition. Open-goal and close high-quality chances received high probabilities, while headers under pressure and difficult long-range shots received low probabilities. This is the main interpretability advantage of the Bayesian Network approach: the models can answer direct conditional-probability questions about football situations, instead of only returning global performance scores.

Conclusion

This project showed that Bayesian Networks can be useful for interpretable xG modeling. StatsBomb xG and Logistic Regression were stronger on several numerical metrics, but the Expert BN produced excellent aggregate calibration and meaningful football-scenario probabilities. The analysis showed that shot geometry, goalkeeper context, and defensive pressure are key drivers of goal probability. The Hill-Climbing BN confirmed that distance, body part, shot type, open-goal context, and first-time shot were locally important around `Goal`. The main limitations are discretization and data sparsity. With more matches, seasons, leagues, and richer tracking data, the learned conditional probability tables and structures could become more accurate and robust.

Links to external resources

Dataset: <https://github.com/statsbomb/open-data>

References

- [1] Constantinou, A. C.; Fenton, N. E.; and Neil, M. 2012. *pi-football: A Bayesian network model for forecasting Association Football match outcomes*. Knowledge-Based Systems, 36, 322–339.
- [2] Mead, J.; O’Hare, A.; and McMenemy, P. 2023. *Expected goals in football: Improving model performance and demonstrating value*. PLOS ONE, 18(4): e0282295.